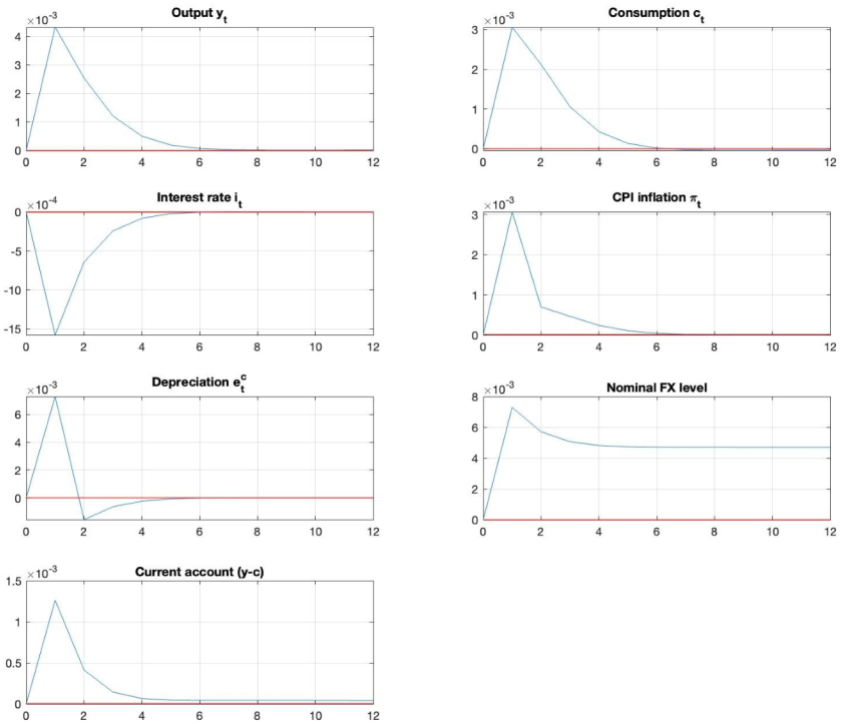
# Quantitative Group Project
# ECON6008 International Money and Finance
# Semester 2 2025
# School of Economics, The University of Sydney

**Group: #21**
**Leonardo Decorte**
**ID: 550905886**

**Q1. Benchmark Foreign Economy (a) Monetary Expansion / Domestic Interest-Rate Shock (+1%)**




A one-time positive monetary policy shock equivalent to a 1% expansion triggers the standard transmission dynamics of a New Keynesian small open economy. On impact, the nominal interest rate drops by about 0.15 percentage points, lowering the real rate and stimulating intertemporal consumption and investment. Consequently, output and consumption are affected, and rise by approximately 0.5% and 0.3%, respectively, before returning to baseline within about six quarters (consistent with the quarterly calibration $\beta = 0.99$).

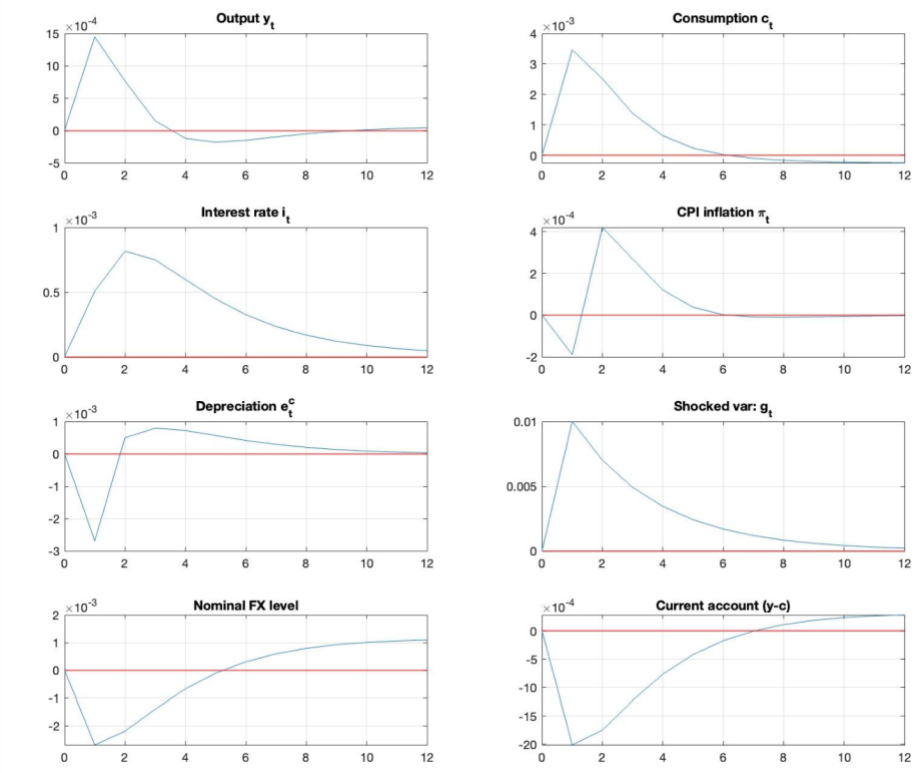
As activity strengthens, CPI inflation responds positively to the impact (+0.3%) as stronger demand and depreciation push prices higher, consistent with the fisher equation ($\pi = i - r$)). Connected to the decrease in the nominal interest rate, the nominal exchange rate (E ) depreciates immediately by about 0.8%, exceeding its long-run response (classic overshooting). This overshooting arises from the uncovered interest parity (UIP) condition: since the domestic rate falls below the foreign rate, investors expect future appreciation, which requires a sharp contemporaneous depreciation. As the policy rule stabilizes expectations, inflation returns to target and the currency gradually appreciates back toward equilibrium.

Reflecting the movements of other variables, especially the exchange rate, the current account briefly moves into surplus as the weaker currency stimulates exports and dampens import demand. Over time, as the interest differential narrows and domestic spending recovers, the surplus narrows. Overall, the shock illustrates the conventional open-economy monetary transmission channel: lower interest rates lead to currency depreciation, a short-term boost in net exports, and a temporary rise in output and prices

before equilibrium is restored through the Taylor rule-type monetary policy that guides the system back to equilibrium, anchoring the inflation expectations and guiding all the variables back to their steady state values.

In the AA–DD model, a monetary expansion lowers domestic interest rates, shifting the AA curve upward as capital outflows and depreciation expectations increase. The resulting weaker currency boosts net exports, moving the economy along the DD curve to a higher level of output and temporary inflation. The DSGE model encapsulates the same qualitative mechanisms within a dynamic, forward-looking framework, aligning with the AA–DD model in terms of direction and transmission channels, yet capturing the gradual, expectation-driven return to equilibrium that the static setup cannot describe.

**(b) Preference (Demand) Shock (+1%)**

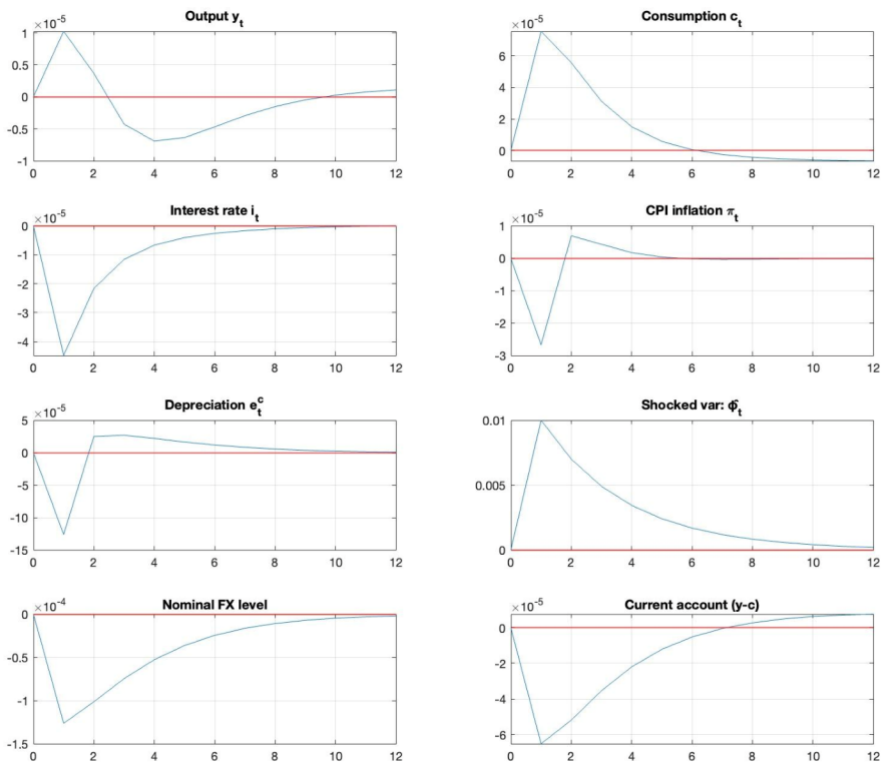



A 1% positive preference (demand) shock increases households' marginal utility of consumption, shifting expenditure forward and stimulating aggregate demand. On impact, output and consumption rise sharply by about 0.14% and 0.35%, respectively, reflecting stronger intertemporal substitution and higher domestic absorption of goods and services. The rise in economic activity and stronger demand pushes inflation up through both domestic price stickiness and imported goods channels. In response to these movements, a subsequent tightening of monetary policy occurs. The central bank, following the Taylor rule, raises the nominal policy rate by about 0.07%, which tightens financial conditions and offsets part of the expansion, gradually bringing inflation back to target.

Via uncovered interest parity (UIP), the higher domestic rate immediately appreciates the nominal exchange rate (≈ −0.25%), leading to a temporary deterioration in net exports and a current-account deficit (≈ −0.20%). As the boom fades and policy normalizes, the deficit gradually closes. The mechanism contrasts with the monetary policy shock: instead of depreciation-driven stimulus, the demand shock triggers a policy-induced appreciation that stabilizes inflation and limits external imbalances.

In AA–DD terms, the higher demand shifts the DD curve rightward, while the induced policy tightening shifts the AA curve downward. Output rises but less than it would in a closed economy, and the current account temporarily deteriorates. Both frameworks predict higher output, prices and currency appreciation.

### (c) Risk-Premium Shock (+1%)




An increase in the risk premium shifts the market beliefs about the small economy. A riskier country issues riskier assets, leading investors to expect higher returns for holding them, and causing a capital outflow. Consistent with UIP, the exchange rate must jump so that expected future movements compensate for the new interest differential. The exchange rate initially appreciates (the growth rate falls), then whipsaws into depreciation as policy and demand respond. This is classic overshooting: sticky prices meet interest-rate differentials, so the FX rate jumps first and goods prices catch up later.

Both households and firms face higher borrowing costs, which in turn reduces their spending plans. At first, output shows a small and short-lived increase, caused by the immediate effects of the exchange-

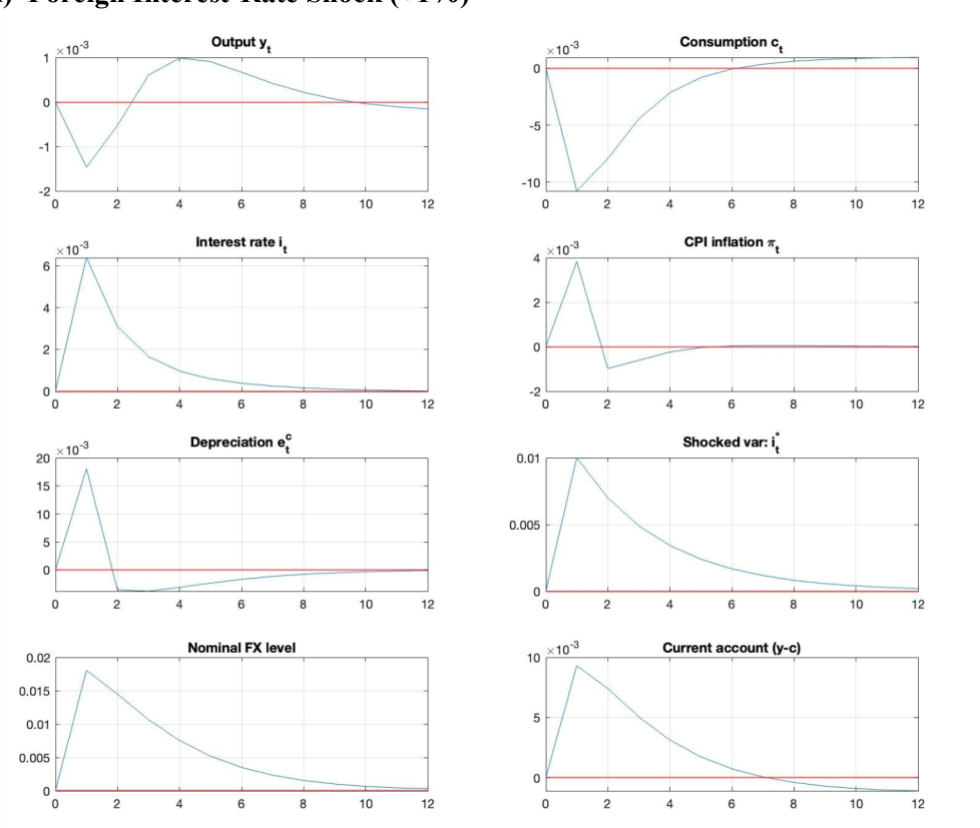
rate movement on relative prices. Soon after, demand weakens, and output drops below its steady-state level. Consumption follows almost the same path: a small bump at the start, then a longer and deeper decline, because households are more sensitive to the cost of credit and income expectations.

The central bank reacts to this downturn by cutting the policy rate to support demand. In the impulse responses, this appears as a sharp fall in the policy rate, followed by a gradual return toward its normal level as the effects of the shock fade and policy adjusts under the Taylor rule. CPI inflation first dips on weak demand, then rebounds modestly above target as the subsequent depreciation raises import prices (the CPI–PPI wedge links CPI to changes in the terms of trade). As activity normalises, inflation returns smoothly to target.

At first glance, the current account deteriorates immediately after the risk-premium shock, as consumption increases more than output in response to the initial, short-lived positive effect of the shock on spending and relative prices. Over time, the current account gradually improves and slightly overshoots its steady state. This recovery reflects the fact that consumption declines more persistently than output, as tighter financial conditions curb domestic demand. In addition, the exchange-rate depreciation that begins around the second quarter supports external competitiveness, boosting net exports. Together, these dynamics drive the current account back toward equilibrium, resulting in a brief surplus.

The mechanism corresponds to a leftward shift of the AA curve: a higher perceived risk discourages capital inflows, causing the currency to depreciate and inflation to rise temporarily. The trade balance partly offsets the fall in output, showing how exchange rate flexibility cushions external financing shocks. However, unlike the smooth adjustment in the AA–DD framework, the New Keynesian model captures the initial opposite movements observed in the IRFs such as the short-lived rise in output and consumption, driven by forward-looking expectations and nominal rigidities before the standard contractionary effects take hold.

### (d) Foreign Interest-Rate Shock (+1%)

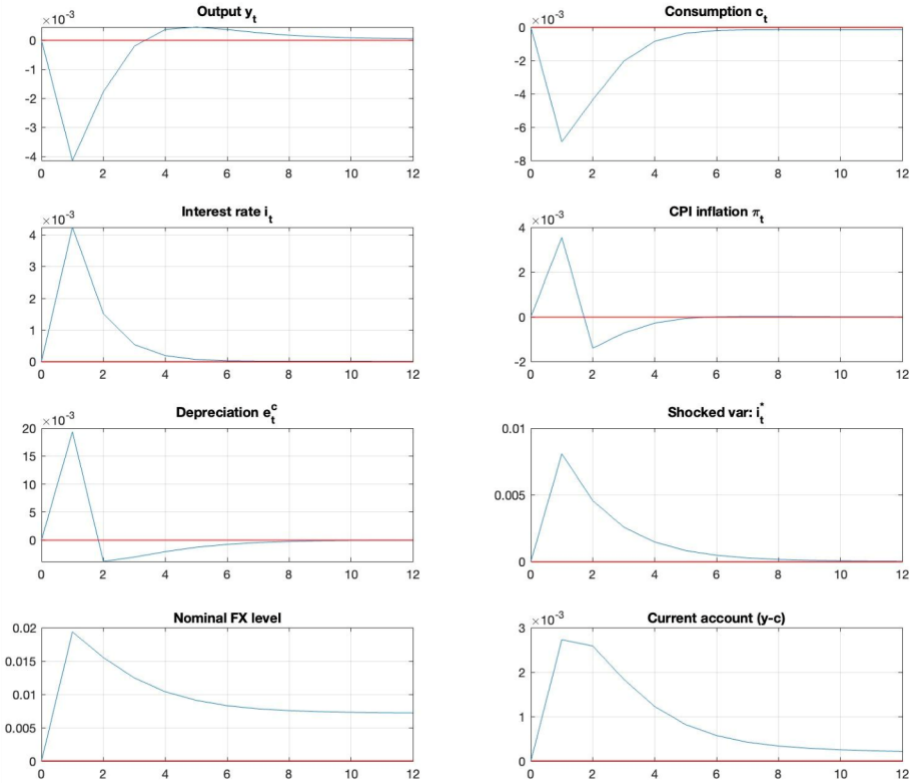



A 1% increase in the foreign interest rate first affects international capital flows, higher foreign returns make domestic assets relatively less attractive, triggering capital outflows and an immediate depreciation of the domestic currency by around 2.0%. This depreciation feeds directly into import prices, pushing inflation up to approximately 0.3–0.4%. To counteract this inflationary pressure, the domestic central bank raises its policy rate by approximately 0.6%, tightening financial conditions. Higher borrowing costs and reduced external financing lead to a decline in output and consumption on impact of about 0.1% and 1.0%, respectively, as investment and household spending contract.

Over time, as demand weakens and inflationary pressures subside, the policy rate gradually returns to its steady-state level. The drop in domestic absorption eases import demand and supports net exports, producing a temporary improvement in the current account. However, as the exchange rate begins to appreciate back, while consumption recovers, the current account surplus narrows again. Eventually, all variables converge smoothly to their steady states as the effects of the external shock fade and monetary policy returns to normal.

In the AA–DD framework, this adjustment corresponds to a leftward shift of both curves: the AA curve shifts left as higher foreign returns reduce capital inflows, and the DD curve shifts left as domestic and external demand weaken. The New Keynesian model, however, captures the initial opposite movements, temporary rebounds in output and consumption, arising from forward-looking expectations and price rigidities before the standard contractionary dynamics dominate.

**Q2. Foreign Block as a Closed-Economy NK Model (Foreign Rate +1%)**




A 1% increase in the foreign interest rate tightens external financial conditions, leading to capital outflows from the domestic economy. This produces an immediate currency depreciation of about 2%, as investors demand higher returns to hold domestic assets. The weaker currency raises import prices, causing CPI inflation to rise by roughly 0.4% on impact, following depreciation. In response, the domestic policy rate is raised by 0.4–0.5%, as the central bank seeks to anchor inflation expectations, and as demand weakens, inflation undershoots the steady state, and the policy rate gradually normalizes.

The tightening in monetary conditions depresses domestic demand and production. Output declines sharply by around 0.4% on impact due to tighter external financial conditions and reduced domestic activity. It then recovers within three quarters, slightly overshooting its steady-state level as the weaker currency improves net exports before returning to equilibrium. Consumption drops by nearly 0.7%, reflecting higher interest rates, reduced real income and consumption smoothing behavior, before gradually returning to its steady state without overshooting.

The nominal exchange rate level, however, remains persistently higher for several quarters, reflecting slow adjustment. The current account improves slightly as lower absorption reduces imports, but it normalises once domestic demand rebounds.

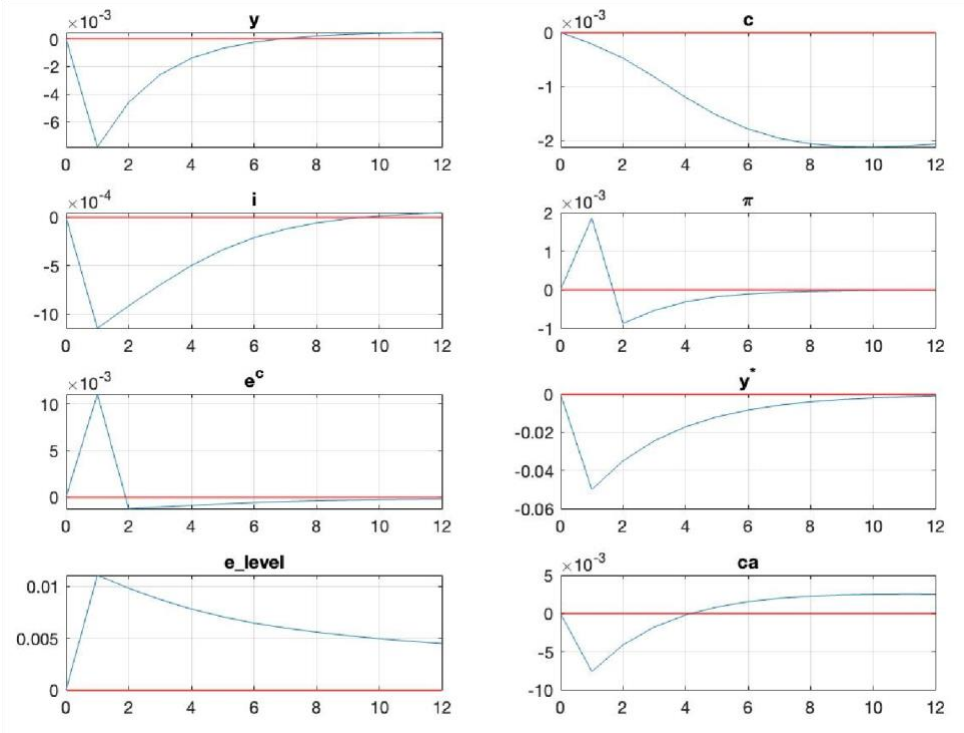
**Comparison with Q1(d)**

Relative to Q1(d), the directions of the exchange rate and current account remain the same, with a currency depreciation and a temporary external surplus. However, in the NK closed foreign economy framework, the output contraction is larger, while the consumption decline is smaller, and the current-account improvement is more limited. This means that the economy stabilizes primarily via exchange-rate and intertemporal mechanisms as nominal rigidities and forward-looking behavior shift the burden of adjustment away from trade volumes (output-driven trade adjustment).

Moreover, the nominal exchange-rate level shows greater persistence, as the interest-rate differential closes only gradually, leaving the domestic currency weaker for several quarters before returning to equilibrium. In both cases, the AA–DD intuition holds, but the New Keynesian framework, by modelling the full adjustment path of variables, provides richer and more valuable insights.

**Q3. GFC-like Shocks and Exchange-Rate Policies**

**(a) Foreign Output Shock under a Floating Exchange Rate (φe = 0)**



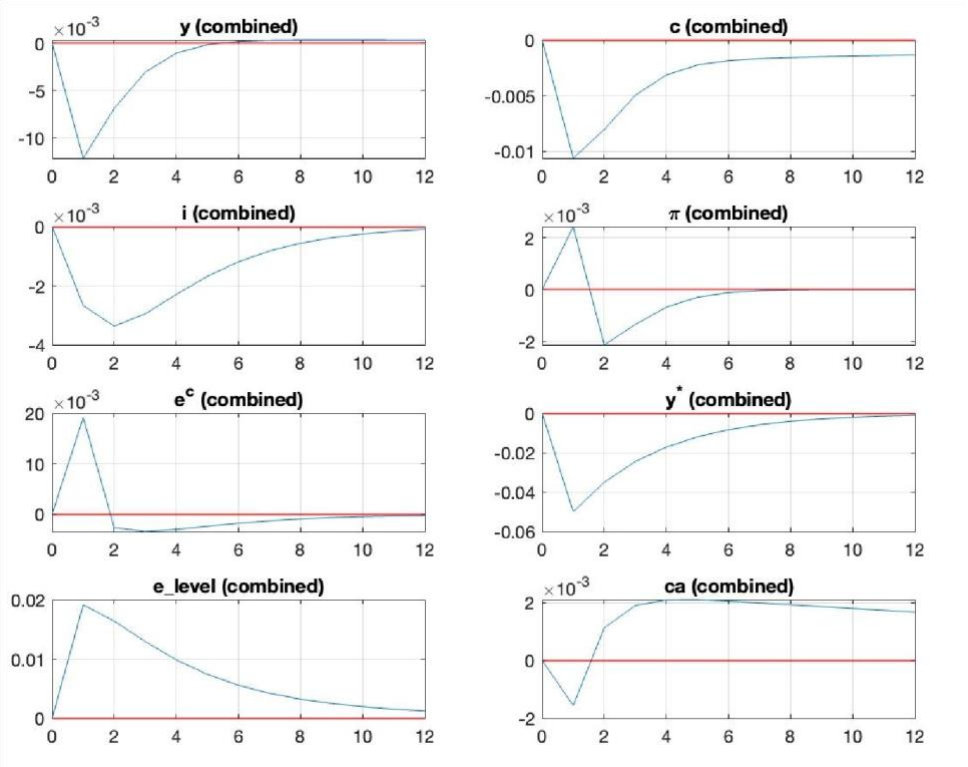

The initial drop in foreign output reduces demand for domestic goods, shifting the goods-market clearing condition curve leftward and causing domestic output to fall. In the model, lower foreign output (y*) reduces export demand through the term ϕy* in the goods-market equilibrium (eq. 2), thereby contracting domestic output and triggering external adjustment. In response, monetary policy eases gradually according to the Taylor rule, leading to a temporary depreciation of the exchange rate. This depreciation briefly supports net exports by improving competitiveness, but as interest rate differentials close, the exchange rate appreciates back toward its steady state. Consequently, the subsequent improvement in the current account stems primarily from reduced domestic absorption rather than sustained export growth.

Indeed, consumption continues to decline because households expect the fall in foreign demand to persist. A key factor behind this pattern is habit formation in the IS equation, which makes households slower to adjust their spending. Weaker trade reduces real income, and the central bank's moderate rate cuts are insufficient to fully offset these contractionary effects. As a result, while output gradually recovers, households bear most of the adjustment through lower consumption.

Inflation initially rises due to import price pass-through from the weaker currency, but as aggregate demand remains subdued, it soon falls below target. Over time, both inflation and output recover as expectations adjust, monetary policy normalizes, and the effects of the foreign output shock dissipate.

Overall, the New Keynesian model preserves the AA–DD intuition—foreign weakness reduces output and depreciates the currency but enhances it by capturing the dynamic path of expectations, monetary policy, and intertemporal decisions, which explain the persistent fall in consumption and the slow normalization of the exchange-rate level.

**(b) Combined Shocks: Foreign Output and Domestic Preference Shocks under a Floating Exchange Rate (φe = 0)**



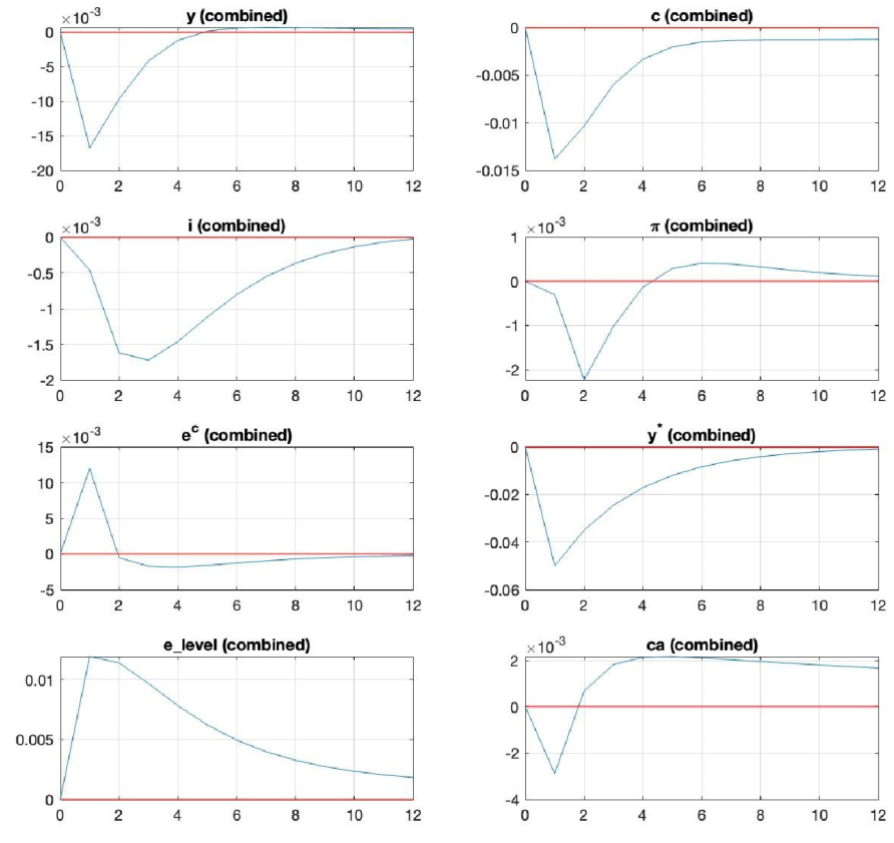

Output drops more steeply than in part (a) because both foreign and domestic demand weaken at the same time: exports fall due to lower external activity, and households cut spending in response to the domestic shock. Monetary policy reacts more aggressively to the combined shocks, sharply cutting interest rates and triggering a strong but temporary depreciation of the exchange rate. This depreciation cushions the fall in income by briefly supporting net exports, while lower real interest rates stimulate intertemporal substitution and help stabilize consumption. Although consumption initially falls, it recovers more quickly than in the foreign-shock-only case. The domestic preference shock, which fades relatively fast, lowers the natural rate of interest and amplifies monetary accommodation; hence, real rates decline enough for households to smooth consumption earlier, even as weak external demand keeps output subdued.

Inflation drops more sharply as both demand components contract, prompting an initial deep rate cut followed by an earlier normalization as recovery expectations strengthen. The exchange rate depreciates on impact but begins appreciating from the second quarter, reflecting the closing interest-rate

differential and improving confidence. The current account adjusts faster and overshoots its steady state, as collapsing domestic demand compresses imports more than exports.

Compared with the single external shock, the combined disturbances shift both the AA and DD curves leftward—the AA curve due to weaker foreign demand and lower capital inflows, and the DD curve due to reduced domestic spending. The overall outcome is a deeper but not more persistent recession, accompanied by faster external adjustment.

**(c) Combined Shocks under a Managed Floating Exchange Rate (φe = 0.80)**



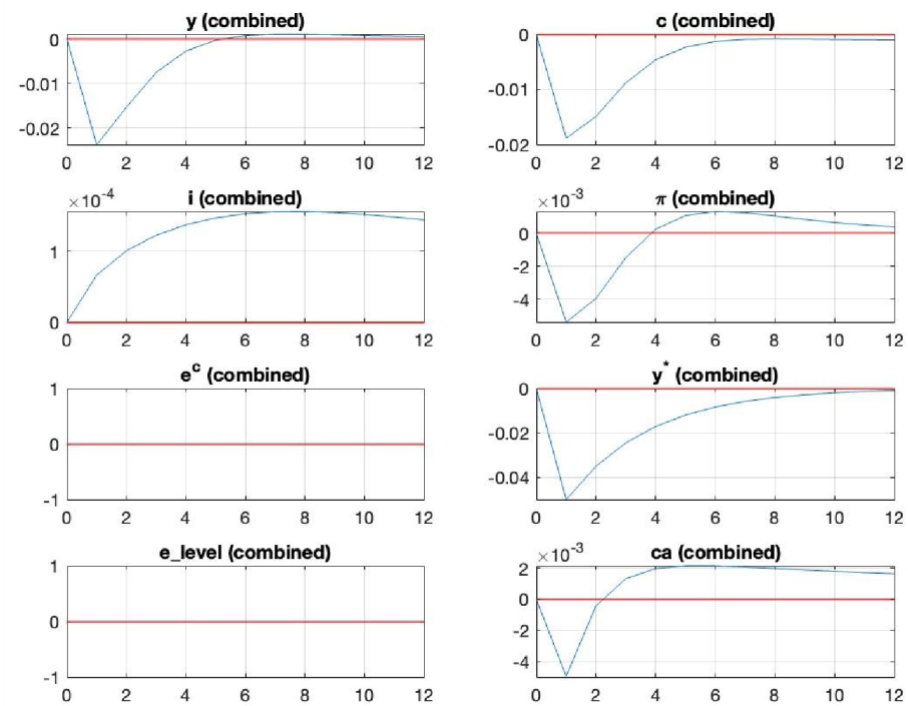

Under a managed float, where the central bank leans against exchange-rate movements, the policy reaction changes substantially as a new term is added to the policy rule, reflecting the exchange-rate's importance for the central bank and smoothing monetary policy response. The result is a smaller depreciation of the domestic currency, although not significantly shorter-lived compared to the pure floating regime. The interest rate still falls about 0,18% after the shocks but by less than under a pure float, reflecting the central bank's effort to contain exchange-rate swings rather than aggressively easing policy.

With the smaller depreciation, inflation drops more sharply but with lower volatility, since there is less exchange-rate pass-through to import prices (Philipps Curve effect). Domestic activity falls more (0.13%) but recovers along a similar path as before, as the smaller depreciation provides less of an

external cushion through exports. Consumption declines but is slightly less persistent, as reduced price and exchange-rate volatility stabilize households' real income expectations. The current account still improves over time but to a lesser extent than under the floating regime, reflecting the reduced expenditure-switching effect from the smaller depreciation.

In AA–DD terms, the reaction to the exchange rate effectively pulls the AA curve back toward its initial position, moderating exchange-rate movements. Consequently, the burden of stabilization falls more heavily on interest rates and domestic demand rather than on the external sector. The managed float thus succeeds in reducing exchange rate and inflation volatility but at the cost of higher real-side variability.

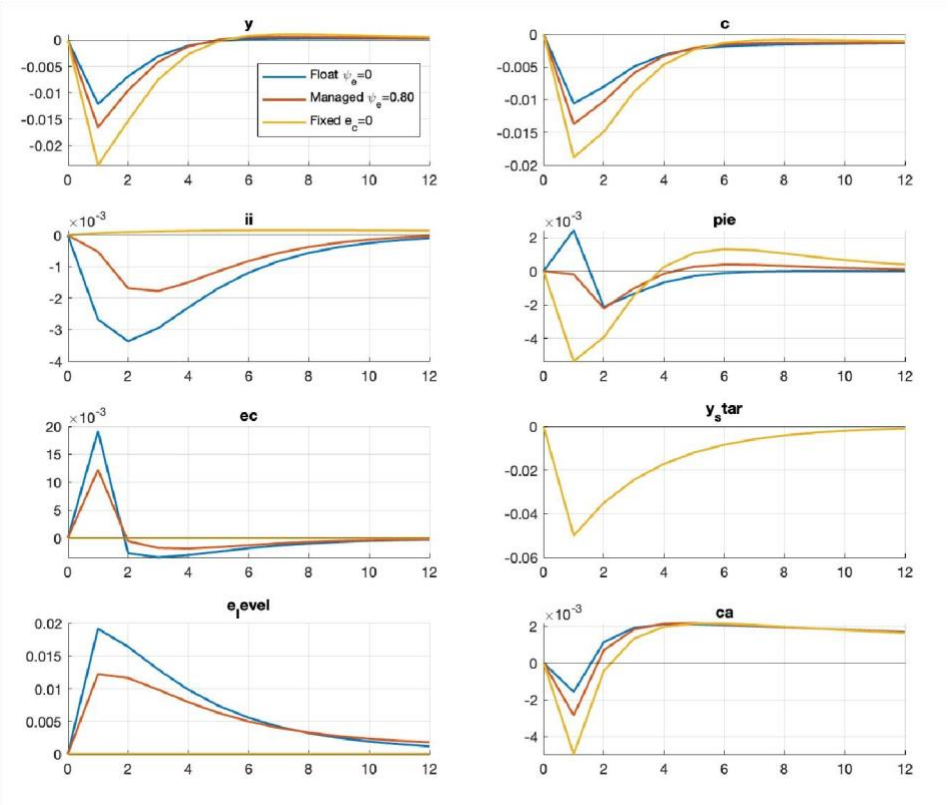
**(d) Combined Shocks under a Fixed Exchange Rate (ec = 0)**




Under a fixed exchange rate, the nominal exchange rate remains constant, leaving the central bank with no ability to use currency adjustments as a stabilization tool. Monetary policy must therefore operate solely through the interest rate to maintain the peg. In the model's impulse responses, both the exchange rate and its level remain perfectly flat, while the policy rate rises slightly. This upward movement reflects the central bank's effort to preserve interest rate parity and defend the peg: lowering rates to stimulate domestic demand would trigger capital outflows and pressure the currency.

CPI inflation reacts slowly at first because import prices are stable, but as the decline in output and demand deepens, inflation falls below target. Once activity begins to normalize, forward-looking firms adjust prices upward, leading inflation to rebound and briefly overshoot its steady state before stabilizing. The absence of exchange-rate flexibility means that all nominal adjustment must take place through domestic prices, resulting in a gradual but oscillatory inflation path.

Output and consumption experience the sharpest and most persistent declines across all regimes (respectively about 2,2% and 1,8%). Without the depreciation mechanism, the economy cannot regain competitiveness to offset the fall in foreign demand, and recovery depends solely on the gradual normalization of external conditions. The current account first worsens (exports drop), then improves quickly, and can turn positive, mainly via import compression as households and firms cut spending.




The three exchange-rate regimes therefore highlight clear trade-offs between real stabilization and nominal stability. Under a floating exchange rate, the sharp but short-lived depreciation supports exports and cushions output, as monetary policy can ease freely. This flexibility stabilizes activity but causes greater volatility in inflation and the exchange rate.

The managed float moderates these fluctuations: tighter policy reduces exchange-rate and price swings but weakens output and consumption, as the smaller depreciation provides less external support. Finally, under a fixed rate, the central bank must raise interest rates to defend the peg, eliminating nominal volatility but amplifying real contraction. Output and consumption fall the most and recover slowly, with the current account improving only through import compression.

Overall, the floating regime best stabilizes real activity, the fixed regime anchors nominal stability, and the managed float offers a costly but balanced middle ground between the two.